OFDM Symbol Recovery and Verification

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2022A3PS0460G

1 System Parameters

- FFT Size: N = 8
- Transmitted symbol (frequency domain): X = [2, -1, 0, 1, 3, 0, -2, 1]
- Channel impulse response: h = [1, 0.5, 0.25]
- Cyclic prefix length: L = 2 (since len(h) = 3)

2 Step 1: Time Domain Symbol Generation

The transmitted symbols are initially given in the frequency domain. To transmit them as a signal, they must be converted into time-domain samples using the IDFT.

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{j2\pi nk/N}$$

Since $N = 8$,

$$x[n] = \frac{1}{8}\sum_{k=0}^{7} X[k]e^{j2\pi nk/8}$$

Calculating manually:

$$x[0] = \frac{2-1+0+1+3+0-2+1}{8} = \frac{4}{8} = 0.5 + 0i$$

$$x[1] = \frac{2 - e^{j\pi/4} + e^{j3\pi/4} + 3e^{j\pi} - 2e^{j3\pi/2} + e^{j7\pi/4}}{8}$$

$$= \frac{2 - (0.7071 + 0.7071i) + (-0.7071 + 0.7071i) - 3 + 2i + (0.7071 - 0.7071i)}{8} = -0.2134{+}0.1616i$$

$$x[2] = \frac{2 - e^{j\pi/2} + e^{j3\pi/2} + 3e^{j2\pi} - 2e^{j3\pi} + e^{j7\pi/2}}{8} = \frac{2 - i - i + 3 + 2 - i}{8} = 0.8750{-}0.3750i$$

$$x[3] = \frac{2 - e^{j3\pi/4} + e^{j9\pi/4} + 3e^{j3\pi} - 2e^{j9\pi/2} + e^{j21\pi/4}}{8}$$

$$= \frac{2 - (-0.7071 + 0.7071i) + (0.7071 + 0.7071i) - 3 - 2i + (-0.7071 - 0.7071i)}{8} = -0.0366{-}0.3384i$$

$$x[4] = \frac{2 - e^{j\pi} + e^{j3\pi} + 3e^{j4\pi} - 2e^{j6\pi} + e^{j7\pi}}{8} = \frac{2 + 1 - 1 + 3 - 2 - 1}{8} = 0.25{+}0i$$

$$x[5] = \frac{2 - e^{j5\pi/4} + e^{j15\pi/4} + 3e^{j5\pi} - 2e^{j15\pi/2} + e^{j35\pi/4}}{8}$$

$$= \frac{2 - (-0.7071 - 0.7071i) + (0.7071 - 0.7071i) - 3 + 2i + (-0.7071 + 0.7071i)}{8} = -0.0366{+}0.3384i$$

$$x[6] = \frac{2 - e^{j3\pi/2} + e^{j9\pi/2} + 3e^{j6\pi} - 2e^{j9\pi} + e^{j21\pi/2}}{8}$$

$$= \frac{2 + i + i + 3 + 2 + i}{8} = 0.8750 + 0.3750i$$

$$x[7] = \frac{2 - e^{j7\pi/4} + e^{j21\pi/4} + 3e^{j7\pi} - 2e^{j21\pi/2} + e^{j49\pi/4}}{8}$$

$$= \frac{2 - (0.7071 - 0.7071i) + (-0.7071 - 0.7071i) - 3 - 2i + (0.7071 + 0.7071i)}{8} = -0.2134{-}0.1616i$$

Therefore,

x = [0.5000 + 0i, -0.2134 + 0.1616i, 0.8750 - 0.3750i, -0.0366 -0.3384i, 0.2500 + 0i, -0.0366 + 0.3384i, 0.8750 + 0.3750i, -0.2134 -0.1616i]

3 Step 2: Cyclic Prefix Addition

The cyclic prefix is added to reduce the effect of inter-symbol interference caused by the channel.

Since $L = 2$, the last two samples of x are copied and added to the beginning.

The last two samples are:

$$x[6] = 0.8750 + 0.3750i$$

$$x[7] = -0.2134 - 0.1616i$$

Therefore,

x_{CP} = [$0.8750{+}0.3750i$, $-0.2134{-}0.1616i$, 0.5000, $-0.2134{+}0.1616i$, $0.8750{-}0.3750i$, $-0.0366{-}0.3384i$, 0.2500, $-0.0366{+}0.3384i$, $0.8750{+}0.3750i$, $-0.2134{-}0.1616i$]

This is the transmitted OFDM signal.

4 Step 3: Channel Convolution

The transmitted signal passes through the channel. This is represented using convolution:

$$y_{CP} = x_{CP} * h$$

Since

$$h = [1, 0.5, 0.25]$$

each output sample is calculated as:

$$y[n] = x_{CP}[n] + 0.5x_{CP}[n-1] + 0.25x_{CP}[n-2]$$

Calculating manually:

$$y[0] = h[0]x_{CP}[0] = 1(0.8750 + 0.3750i) = 0.8750 + 0.3750i$$

$$y[1] = h[0]x_{CP}[1] + h[1]x_{CP}[0] = 1(-0.2134 - 0.1616i) + 0.5(0.8750 + 0.3750i) = 0.2241 + 0.0259i$$

$$y[2] = h[0]x_{CP}[2] + h[1]x_{CP}[1] + h[2]x_{CP}[0] = 1(0.5000) + 0.5(-0.2134 - 0.1616i) + 0.25(0.8750 + 0.3750i)$$

$$= 0.6121 + 0.0129i$$

$$y[3] = h[0]x_{CP}[3] + h[1]x_{CP}[2] + h[2]x_{CP}[1]$$

$$= 1(-0.2134 + 0.1616i) + 0.5(0.5000) + 0.25(-0.2134 - 0.1616i)$$

$$= -0.0167 + 0.1212i$$

$$y[4] = h[0]x_{CP}[4] + h[1]x_{CP}[3] + h[2]x_{CP}[2]$$

$$= 1(0.8750 - 0.3750i) + 0.5(-0.2134 + 0.1616i) + 0.25(0.5000)$$

$$= 0.8933 - 0.2942i$$

$$y[5] = h[0]x_{CP}[5] + h[1]x_{CP}[4] + h[2]x_{CP}[3]$$

$$= 1(-0.0366 - 0.3384i) + 0.5(0.8750 - 0.3750i) + 0.25(-0.2134 + 0.1616i)$$

$$= 0.3475 - 0.4855i$$

$$y[6] = h[0]x_{CP}[6] + h[1]x_{CP}[5] + h[2]x_{CP}[4]$$

$$= 1(0.2500) + 0.5(-0.0366 - 0.3384i) + 0.25(0.8750 - 0.3750i)$$

$$= 0.4504 - 0.2629i$$

$$y[7] = h[0]x_{CP}[7] + h[1]x_{CP}[6] + h[2]x_{CP}[5]$$

$$= 1(-0.0366 + 0.3384i) + 0.5(0.2500) + 0.25(-0.0366 - 0.3384i)$$

$$= 0.0792 + 0.2538i$$

$$y[8] = h[0]x_{CP}[8] + h[1]x_{CP}[7] + h[2]x_{CP}[6]$$

$$= 1(0.8750 + 0.3750i) + 0.5(-0.0366 + 0.3384i) + 0.25(0.2500)$$

$$= 0.9192 + 0.5442i$$

$$y[9] = h[0]x_{CP}[9] + h[1]x_{CP}[8] + h[2]x_{CP}[7]$$

$$= 1(-0.2134 - 0.1616i) + 0.5(0.8750 + 0.3750i) + 0.25(-0.0366 + 0.3384i)$$

$$= 0.2150 + 0.1105i$$

$$y[10] = h[1]x_{CP}[9] + h[2]x_{CP}[8] = 0.5(-0.2134 - 0.1616i) + 0.25(0.8750 + 0.3750i) = 0.1121 + 0.0129i$$

$$y[11] = h[2]x_{CP}[9] = 0.25(-0.2134 - 0.1616i) = -0.0533 - 0.0404i$$

Therefore,

$\mathrm{y}_{CP} = [0.8750+0.3750i,\ 0.2241+0.0259i,\ 0.6121+0.0129i,\ -0.0167+0.1212i,\ 0.8933-0.2942i,\ 0.3475-0.4855i,\ 0.4504-0.2629i,\ 0.0792+0.2538i,\ 0.9192+0.5442i,\ 0.2150+0.1105i,\ 0.1121+0.0129i,\ -0.0533-0.0404i]$

5 Step 4: CP Removal

At the receiver, the cyclic prefix is removed because it was only added for protection against channel delay.

Since $L = 2$, the first two samples are removed, and the next $N = 8$ samples are kept.

$\text{y}_{received} = [0.6121 + 0.0129i,\ -0.0167 + 0.1212i,\ 0.8933 - 0.2942i,\ 0.3475 - 0.4855i,\ 0.4504 - 0.2629i,\ 0.0792 + 0.2538i,\ 0.9192 + 0.5442i,\ 0.2150 + 0.1105i]$

6 Step 5: Frequency-Domain Processing

The receiver applies FFT to convert the received time-domain signal back into the frequency domain.

6.1 Compute DFT of Received Signal

The frequency-domain signal is obtained using the DFT to convert the received time-domain signal:

$$Y[k] = \sum_{n=0}^{7} y[n] e^{-j2\pi kn/8}$$

The DFT is calculated as:

$$Y[k] = \sum_{n=0}^{7} y[n] e^{-j2\pi kn/8}$$

For $k = 0$:

$$Y[0] = \sum_{n=0}^{7} y[n]$$

$$Y[0] = (0.6121 + 0.0129i) + (-0.0167 + 0.1212i) + (0.8933 - 0.2942i)$$

$$+(0.3475 - 0.4855i) + (0.4504 - 0.2629i) + (0.0792 + 0.2538i)$$

$$+(0.9192 + 0.5442i) + (0.2150 + 0.1105i)$$

$$Y[0] = 3.5000 + 0i$$

For $k = 1$:

$$Y[1] = \sum_{n=0}^{7} y[n] e^{-j2\pi n/8}$$

$$Y[1] = -1.3536 + 0.6036i$$

For $k = 2$:

$$Y[2] = \sum_{n=0}^{7} y[n]e^{-j2\pi(2)n/8}$$

$$Y[2] = 0 + 0i$$

For $k = 3$:

$$Y[3] = \sum_{n=0}^{7} y[n]e^{-j2\pi(3)n/8}$$

$$Y[3] = 0.6464 - 0.1036i$$

For $k = 4$:

$$Y[4] = \sum_{n=0}^{7} y[n]e^{-j2\pi(4)n/8}$$

$$Y[4] = \sum_{n=0}^{7} y[n]e^{-j\pi n}$$

Since:

$$e^{-j\pi n} = (-1)^n$$

$$Y[4] = y[0] - y[1] + y[2] - y[3] + y[4] - y[5] + y[6] - y[7]$$

$$Y[4] = 2.2500 + 0i$$

For $k = 5$:

$$Y[5] = \sum_{n=0}^{7} y[n]e^{-j2\pi(5)n/8}$$

$$Y[5] = 0 + 0i$$

For $k = 6$:

$$Y[6] = \sum_{n=0}^{7} y[n]e^{-j2\pi(6)n/8}$$

$$Y[6] = -1.5000 - 1.0000i$$

For $k = 7$:

$$Y[7] = \sum_{n=0}^{7} y[n] e^{-j2\pi(7)n/8}$$

$$Y[7] = 1.3536 + 0.6036i$$

Therefore,
Y = [3.5000 + 0i, -1.3536 + 0.6036i, 0 + 0i, 0.6464 - 0.1036i, 2.2500 + 0i, 0 + 0i, -1.5000 - 1.0000i, 1.3536 + 0.6036i]

6.2 Compute Channel Frequency Response

The channel impulse response is:

$$h = [1, 0.5, 0.25]$$

Since the FFT size is $N = 8$, the channel is zero-padded to length 8:

$$h_{padded} = [1, 0.5, 0.25, 0, 0, 0, 0, 0]$$

The channel frequency response is obtained using the DFT:

$$H[k] = \sum_{n=0}^{7} h[n] e^{-j2\pi kn/8}$$

Since only the first three values of h are non-zero:

$$H[k] = 1 + 0.5e^{-j2\pi k/8} + 0.25e^{-j2\pi k(2)/8}$$

Calculating manually:

$$H[0] = 1 + 0.5e^{-j2\pi(0)/8} + 0.25e^{-j2\pi(0)(2)/8}$$

$$= 1 + 0.5 + 0.25 = 1.7500 + 0i$$

$$H[1] = 1 + 0.5e^{-j2\pi(1)/8} + 0.25e^{-j2\pi(1)(2)/8}$$

$$= 1 + 0.5e^{-j\pi/4} + 0.25e^{-j\pi/2}$$

$$= 1 + 0.5(0.7071 - 0.7071i) + 0.25(0 - 1i)$$

$$= 1 + 0.3536 - 0.3536i - 0.25i$$

$$H[1] = 1.3536 - 0.6036i$$

$$H[2] = 1 + 0.5e^{-j2\pi(2)/8} + 0.25e^{-j2\pi(2)(2)/8}$$

$$= 1 + 0.5e^{-j\pi/2} + 0.25e^{-j\pi}$$

$$= 1 + 0.5(0 - 1i) + 0.25(-1)$$

$$= 1 - 0.5i - 0.25$$

$$H[2] = 0.7500 - 0.5000i$$

$$H[3] = 1 + 0.5e^{-j2\pi(3)/8} + 0.25e^{-j2\pi(3)(2)/8}$$

$$= 1 + 0.5e^{-j3\pi/4} + 0.25e^{-j3\pi/2}$$

$$= 1 + 0.5(-0.7071 - 0.7071i) + 0.25(0 + 1i)$$

$$= 1 - 0.3536 - 0.3536i + 0.25i$$

$$H[3] = 0.6464 - 0.1036i$$

$$H[4] = 1 + 0.5e^{-j2\pi(4)/8} + 0.25e^{-j2\pi(4)(2)/8}$$

$$= 1 + 0.5e^{-j\pi} + 0.25e^{-j2\pi}$$

$$= 1 + 0.5(-1) + 0.25(1)$$

$$= 1 - 0.5 + 0.25$$

$$H[4] = 0.7500 + 0i$$

$$H[5] = 1 + 0.5e^{-j2\pi(5)/8} + 0.25e^{-j2\pi(5)(2)/8}$$

$$= 1 + 0.5e^{-j5\pi/4} + 0.25e^{-j5\pi/2}$$

$$= 1 + 0.5(-0.7071 + 0.7071i) + 0.25(0 - 1i)$$

$$= 1 - 0.3536 + 0.3536i - 0.25i$$

$$H[5] = 0.6464 + 0.1036i$$

$$H[6] = 1 + 0.5e^{-j2\pi(6)/8} + 0.25e^{-j2\pi(6)(2)/8}$$

$$= 1 + 0.5e^{-j3\pi/2} + 0.25e^{-j3\pi}$$

$$= 1 + 0.5(0 + 1i) + 0.25(-1)$$

$$= 1 + 0.5i - 0.25$$

$$H[6] = 0.7500 + 0.5000i$$

$$H[7] = 1 + 0.5e^{-j2\pi(7)/8} + 0.25e^{-j2\pi(7)(2)/8}$$

$$= 1 + 0.5e^{-j7\pi/4} + 0.25e^{-j7\pi/2}$$

$$= 1 + 0.5(0.7071 + 0.7071i) + 0.25(0 + 1i)$$

$$= 1 + 0.3536 + 0.3536i + 0.25i$$

$$H[7] = 1.3536 + 0.6036i$$

Therefore,

H = [1.7500 + 0i, 1.3536 - 0.6036i, 0.7500 - 0.5000i, 0.6464 - 0.1036i, 0.7500 + 0i, 0.6464 + 0.1036i, 0.7500 + 0.5000i, 1.3536 + 0.6036i]

6.3 Equalization

In the frequency domain, the received signal is related to the transmitted signal by:

$$Y[k] = X[k]H[k]$$

Therefore, to recover the transmitted symbol:

$$\hat{X}[k] = \frac{Y[k]}{H[k]}$$

Using zero-forcing equalization:

$$\hat{X} = Y./H$$

Calculating manually:

$$\hat{X}[0] = \frac{Y[0]}{H[0]} = \frac{3.5000 + 0i}{1.7500 + 0i}$$

$$\hat{X}[0] = 2$$

$$\hat{X}[1] = \frac{Y[1]}{H[1]} = \frac{-1.3536 + 0.6036i}{1.3536 - 0.6036i}$$

Multiplying numerator and denominator by the conjugate of the denominator:

$$\hat{X}[1] = \frac{(-1.3536 + 0.6036i)(1.3536 + 0.6036i)}{(1.3536 - 0.6036i)(1.3536 + 0.6036i)}$$

$$= \frac{(-1.3536)^2 - (0.6036)^2}{(1.3536)^2 + (0.6036)^2}$$

$$= \frac{-2.1960}{2.1960}$$

$$\hat{X}[1] = -1$$

$$\hat{X}[2] = \frac{Y[2]}{H[2]} = \frac{0 + 0i}{0.7500 - 0.5000i}$$

$$\hat{X}[2] = 0$$

$$\hat{X}[3] = \frac{Y[3]}{H[3]} = \frac{0.6464 - 0.1036i}{0.6464 - 0.1036i}$$

$$\hat{X}[3] = 1$$

$$\hat{X}[4] = \frac{Y[4]}{H[4]} = \frac{2.2500 + 0i}{0.7500 + 0i}$$

$$\hat{X}[4] = 3$$

$$\hat{X}[5] = \frac{Y[5]}{H[5]} = \frac{0 + 0i}{0.6464 + 0.1036i}$$

$$\hat{X}[5] = 0$$

$$\hat{X}[6] = \frac{Y[6]}{H[6]} = \frac{-1.5000 - 1.0000i}{0.7500 + 0.5000i}$$

Multiplying numerator and denominator by the conjugate:

$$\hat{X}[6] = \frac{(-1.5000 - 1.0000i)(0.7500 - 0.5000i)}{(0.7500 + 0.5000i)(0.7500 - 0.5000i)}$$

$$= \frac{-1.6250 + 0i}{0.8125}$$

$$\hat{X}[6] = -2$$

$$\hat{X}[7] = \frac{Y[7]}{H[7]} = \frac{1.3536 + 0.6036i}{1.3536 + 0.6036i}$$

$$\hat{X}[7] = 1$$

Therefore,

$$\hat{X} = [2, -1, 0, 1, 3, 0, -2, 1]$$

The recovered symbol is the same as the transmitted symbol:

$$\hat{X} = X$$

Hence, perfect symbol recovery is achieved.

7 Discussion of Results

Perfect symbol recovery is achieved in this OFDM system, as the recovered transmitted symbols match the original transmitted symbols:

$$\hat{X} = X$$

This successful recovery is due to the following reasons:

- The cyclic prefix length is sufficient for the given channel. Since the channel impulse response has length 3, a cyclic prefix of length 2 ensures that inter-symbol interference (ISI) is avoided. This allows the linear convolution of the channel to behave like circular convolution, which is required for proper OFDM operation.
- The channel frequency response does not contain any zero values. Since all $H[k] \neq 0$, the division operation during zero-forcing equalization is valid, allowing accurate recovery of the transmitted symbols.

In practical systems, additional considerations are required:

- A longer cyclic prefix may be used to handle more severe channel delay spreads.
- Minimum Mean Square Error (MMSE) equalization can be used instead of zero-forcing to improve performance in the presence of noise.
- Channel coding techniques can be applied to improve reliability and error correction.

Thus, this implementation demonstrates ideal OFDM symbol recovery under perfect channel conditions.

Contents


```
clc;
clear;
close all;

% ==============================
% OFDM Symbol Recovery Assignment
% Name: Parth Nair
% Student ID: 2022A3PS0460G
% ==============================

disp('OFDM Symbol Recovery and Verification');
disp('Name: Parth Nair');
disp('Student ID: 2022A3PS0460G');
```

```
OFDM Symbol Recovery and Verification
Name: Parth Nair
Student ID: 2022A3PS0460G
```

Step 1: System Parameters

```
N = 8;
X = [2 -1 0 1 3 0 -2 1];
h = [1 0.5 0.25];

L = length(h) - 1;

disp(' ');
disp('Step 1: System Parameters');
disp('FFT size N = ');
disp(N);

disp('Transmitted frequency-domain symbol X = ');
disp(X);

disp('Channel impulse response h = ');
disp(h);

disp('Cyclic prefix length L = ');
disp(L);

disp('Name: Parth Nair');
disp('Student ID: 2022A3PS0460G');
```

```
Step 1: System Parameters
FFT size N =
     8

Transmitted frequency-domain symbol X =
     2    -1     0     1     3     0    -2     1

Channel impulse response h =
    1.0000    0.5000    0.2500

Cyclic prefix length L =
     2

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```

Step 2: Time-Domain Symbol Generation using IFFT

```
x = ifft(X);

disp(' ');
disp('Step 2: Time-domain OFDM symbol x = ifft(X)');
disp(x.');

disp('Name: Parth Nair');
disp('Student ID: 2022A3PS0460G');
```

```
Step 2: Time-domain OFDM symbol x = ifft(X)
   0.5000 + 0.0000i
  -0.2134 + 0.1616i
   0.8750 - 0.3750i
  -0.0366 - 0.3384i
   0.2500 + 0.0000i
  -0.0366 + 0.3384i
   0.8750 + 0.3750i
  -0.2134 - 0.1616i

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Student ID: 2022A3PS0460G
```

Step 3: Add Cyclic Prefix

```
x_cp = [x(end-L+1:end) x];

disp(' ');
disp('Step 3: Transmitted OFDM signal with cyclic prefix x_cp');
disp(x_cp.');

disp('Name: Parth Nair');
disp('Student ID: 2022A3PS0460G');
```

```
Step 3: Transmitted OFDM signal with cyclic prefix x_cp
   0.8750 + 0.3750i
  -0.2134 - 0.1616i
   0.5000 + 0.0000i
  -0.2134 + 0.1616i
   0.8750 - 0.3750i
  -0.0366 - 0.3384i
   0.2500 + 0.0000i
  -0.0366 + 0.3384i
   0.8750 + 0.3750i
  -0.2134 - 0.1616i

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Student ID: 2022A3PS0460G
```

Step 4: Pass Signal Through Channel

```
y_cp = conv(x_cp, h);

disp(' ');
disp('Step 4: Output after channel convolution y_cp = conv(x_cp, h)');
disp(y_cp.');

disp('Name: Parth Nair');
disp('Student ID: 2022A3PS0460G');
```

```
Step 4: Output after channel convolution y_cp = conv(x_cp, h)
   0.8750 + 0.3750i
   0.2241 + 0.0259i
   0.6121 + 0.0129i
  -0.0167 + 0.1212i
   0.8933 - 0.2942i
   0.3475 - 0.4855i
   0.4504 - 0.2629i
   0.0792 + 0.2538i
   0.9192 + 0.5442i
   0.2150 + 0.1105i
   0.1121 + 0.0129i
  -0.0533 - 0.0404i

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```

Step 5: Remove Cyclic Prefix

```
y_received = y_cp(L+1:L+N);

disp(' ');
disp('Step 5: Received signal after removing cyclic prefix');
disp(y_received.');

disp('Name: Parth Nair');
disp('Student ID: 2022A3PS0460G');
```

```
Step 5: Received signal after removing cyclic prefix
   0.6121 + 0.0129i
  -0.0167 + 0.1212i
   0.8933 - 0.2942i
   0.3475 - 0.4855i
   0.4504 - 0.2629i
   0.0792 + 0.2538i
   0.9192 + 0.5442i
   0.2150 + 0.1105i

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```

Step 6: FFT of Received Signal

```
Y = fft(y_received);

disp(' ');
disp('Step 6: Frequency-domain received signal Y = fft(y_received)');
disp(Y.');

disp('Name: Parth Nair');
disp('Student ID: 2022A3PS0460G');
```

```
Step 6: Frequency-domain received signal Y = fft(y_received)
   3.5000 - 0.0000i
  -1.3536 + 0.6036i
   0.0000 + 0.0000i
   0.6464 - 0.1036i
   2.2500 + 0.0000i
  -0.0000 + 0.0000i
  -1.5000 - 1.0000i
   1.3536 + 0.6036i

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Student ID: 2022A3PS0460G
```

Step 7: Channel Frequency Response

```
h_padded = [h zeros(1, N - length(h))];
H = fft(h_padded);

disp(' ');
disp('Step 7: Zero-padded channel h_padded');
disp(h_padded.');

disp('Channel frequency response H = fft(h_padded)');
disp(H.');

disp('Name: Parth Nair');
disp('Student ID: 2022A3PS0460G');
```

```
Step 7: Zero-padded channel h_padded
    1.0000
    0.5000
    0.2500
         0
         0
         0
         0
         0

Channel frequency response H = fft(h_padded)
   1.7500 + 0.0000i
   1.3536 - 0.6036i
   0.7500 - 0.5000i
   0.6464 - 0.1036i
   0.7500 + 0.0000i
   0.6464 + 0.1036i
   0.7500 + 0.5000i
   1.3536 + 0.6036i

Name: Parth Nair
Student ID: 2022A3PS0460G
```

Step 8: Equalization

```
X_hat = Y ./ H;

disp(' ');
disp('Step 8: Recovered transmitted symbols X_hat = Y ./ H');
disp(X_hat.');

disp('Name: Parth Nair');
disp('Student ID: 2022A3PS0460G');
```

```
Step 8: Recovered transmitted symbols X_hat = Y ./ H
   2.0000 - 0.0000i
  -1.0000 + 0.0000i
   0.0000 + 0.0000i
   1.0000 + 0.0000i
   3.0000 + 0.0000i
  -0.0000 + 0.0000i
  -2.0000 + 0.0000i
   1.0000 - 0.0000i

Name: Parth Nair
Student ID: 2022A3PS0460G
```

Step 9: Verification

```
disp(' ');
disp('Step 9: Verification');

disp('Original X:');
disp(X.');

disp('Recovered X_hat rounded:');
disp(round(real(X_hat)).');

if isequal(round(real(X_hat)), X)
    disp('Result: Recovered symbols match transmitted symbols.');
else
    disp('Result: Recovered symbols do not match transmitted symbols.');
end

disp('Name: Parth Nair');
disp('Student ID: 2022A3PS0460G');
```

```
Step 9: Verification
Original X:
     2
    -1
     0
     1
     3
     0
    -2
     1

Recovered X_hat rounded:
     2
    -1
     0
     1
     3
     0
    -2
     1

Result: Recovered symbols match transmitted symbols.
Name: Parth Nair
Student ID: 2022A3PS0460G
```